

Colour-Blind-Safe Plots

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Introduction

Roughly 8 % of men of European descent and about 0.5 % of women have some form of colour-vision deficiency. Red-green deficiency (deuteranopia and protanopia) is by far the most common; blue-yellow (tritanopia) is rarer but still present in any large readership. A figure that relies on red-versus-green as its only contrast is unreadable for a meaningful fraction of every audience, including journal reviewers. Designing for colour-blind safety is therefore not optional politeness — it is a basic requirement of accessible scientific communication, increasingly mandated by journal style guides and by web-accessibility standards (WCAG 2.x AA).

Prerequisites

A working knowledge of `ggplot2`'s colour and fill scales, basic familiarity with palette functions, and an understanding that “colour” includes both qualitative palettes (categorical groups) and sequential / diverging palettes (ordered or signed values).

Theory

Safe design strategies fall into three categories:

- **Perceptually uniform palettes:** `viridis`, `cividis`, `magma`, and `inferno` are designed so that equal steps in the colour space correspond to equal steps in perceived brightness, even under simulated colour-blindness. They handle the continuous case well.
- **Hand-curated qualitative palettes:** `RColorBrewer`'s `Set2`, `Dark2`, and `Paired` are explicitly tested for colour-blind safety up to a certain number of categories (six to eight); beyond that, distinguishability degrades.
- **Redundant encoding:** pair colour with shape, linetype, line weight, or a direct text label so a reader who cannot distinguish the colours can still tell the groups apart from a second visual channel.

Simulation tools such as `colorBlindness::cvdPlot()` and `dichromat` re-render any `ggplot` under deuteranopia, protanopia, and tritanopia simulations; treat them as the colour equivalent of a spell-checker run before submission.

Assumptions

No statistical assumptions; the only constraint is that the figure's information content survives loss of one colour channel.

R Implementation

```
library(ggplot2); library(viridis); library(colorBlindness)

# Unsafe: default red-green
p_bad <- ggplot(mtcars, aes(wt, mpg, colour = factor(cyl))) +
  geom_point(size = 3) +
  scale_colour_manual(values = c("red", "orange", "green")) +
  theme_minimal()
```

```
# Safe: viridis + shape redundancy
p_good <- ggplot(mtcars, aes(wt, mpg, colour = factor(cyl), shape = factor(cyl))) +
  geom_point(size = 3) +
  scale_colour_viridis_d(option = "D", end = 0.85) +
  theme_minimal()

cvdPlot(p_good) # simulates deuteranopia, protanopia, tritanopia
```

Output & Results

`cvdPlot()` returns a four-panel composite showing the figure under normal vision, deuteranopia, protanopia, and tritanopia. A safely designed plot remains interpretable in all four panels; an unsafe one collapses to indistinguishable groups in at least one.

Interpretation

A reporting sentence (methods or figure-preparation note): “All figures use the `viridis` palette for ordered fills and `RColorBrewer::Set2` for qualitative colours; categorical groups are encoded redundantly with shape so colour-blind readers can distinguish them. Plots were simulated under deuteranopia and protanopia (`colorBlindness::cvdPlot`) before submission.” Mentioning the accessibility check pre-empts reviewer requests to redo the figures.

Practical Tips

- Default to `viridis` or `cividis` for continuous scales; both are perceptually uniform and remain ordered under all common colour-blindness simulations.
- For categorical groups, pair colour with shape (`shape = group`) or with linetype (`linetype = group`); colour alone is insufficient.
- Avoid red-on-green for success/failure annotations; use blue-on-orange or a categorical palette like `Okabe-Ito`.
- Use `colorBlindness::cvdPlot()` (or the `Color Oracle` desktop tool) on every figure before submission; simulating colour-vision deficiencies takes seconds and catches problems that the human eye cannot.
- Beyond colour, ensure adequate contrast (WCAG AA minimum 4.5:1 for body text), avoid pure red or pure green for thin lines, and keep font sizes large enough to remain legible in greyscale photocopies.
- For network or flow diagrams, prefer position and topology to colour as the primary information channel; colour should be reinforcement, not the only signal.

R Packages Used

`viridis` for perceptually uniform sequential palettes, `RColorBrewer` for qualitative and diverging palettes (`brewer.pal`, `display.brewer.all(colorblindFriendly = TRUE)`), `colorBlindness` for `cvdPlot()` simulations, and `paletteer` for unified access to dozens of curated palettes from across the R ecosystem.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- `ggplot2` grammar and multi-panel layouts